

A

B

C

D

E

F

A

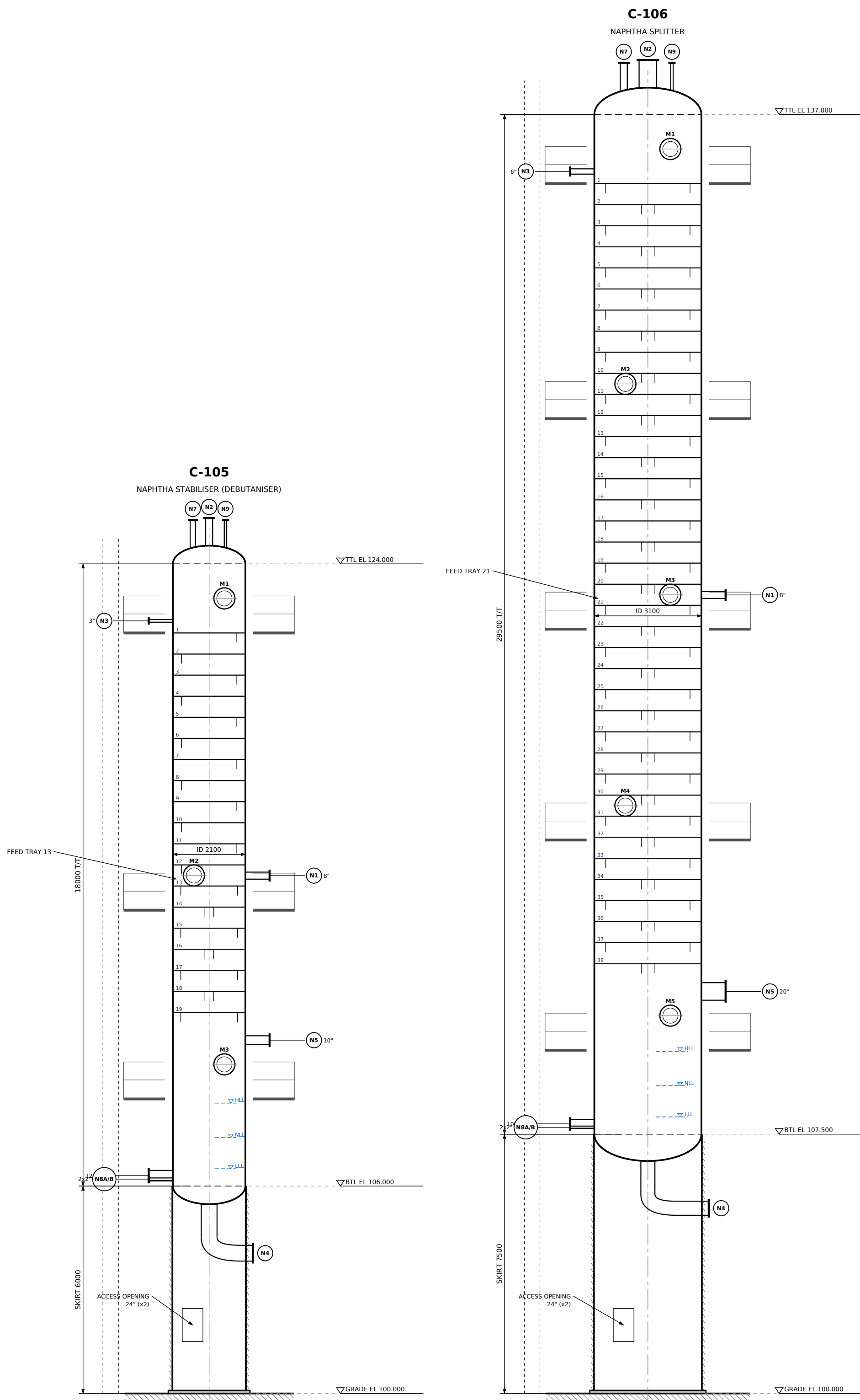
B

C

D

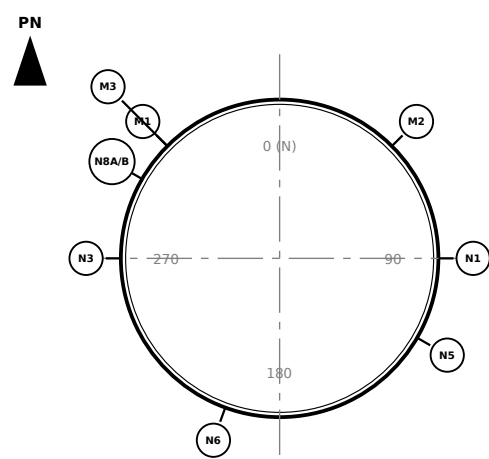
E

F

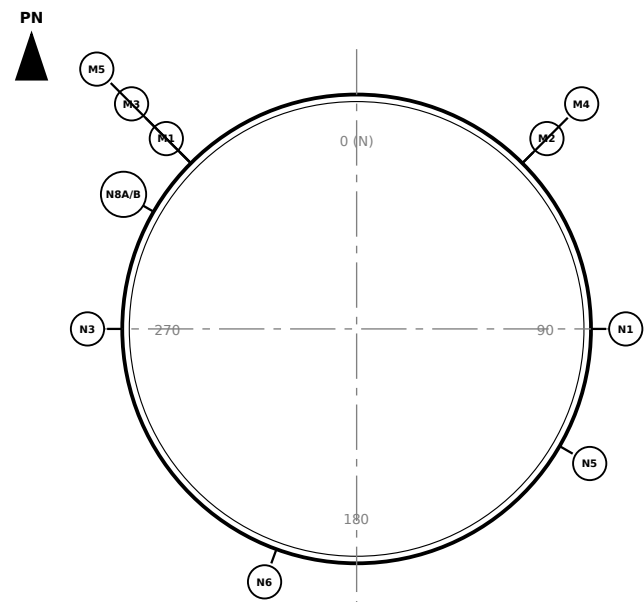


C-105 NOZZLE SCHEDULE						
MARK	QTY	SIZE	RATING	LOCATION	ORIENT	SERVICE
N1	1	8"	300# RF	EL 114.980	90	Feed (tray 13)
N2	1	8"	300# RF	TOP HEAD	0	Overhead vapour to A-106
N3	1	3"	300# RF	EL 122.350	270	Reflux
N4	1	18"	300# RF	BTM HEAD	0	Reboiler feed to E-116 (kettle)
N5	1	10"	300# RF	EL 110.220	120	Reboiler return from E-116 (kettle)
N6	1	12"	300# RF	EL 106.300	200	Stabilised naphtha (bottoms) to E-114/C-106
N7	1	6"	300# RF	TOP HEAD	0	PSV-1005 relief (N)
NBA/B	2	2"	300# RF	EL 106.200	300	LT / LG bridle
N9	1	2"	300# RF	TOP HEAD	0	Vent / steam-out
M1	1	24"	300# RF	EL 123.000	0	Manway 24"
M2	1	24"	300# RF	EL 114.985	0	Manway 24"
M3	1	24"	300# RF	EL 109.530	0	Manway 24"
SECTION						
Top (rectifying)	1-12	FLV	0.138	u flood m/s	D calc m	D sel. m
Bottom (stripping)	13-19	FLV	1.554	0.08	2.04	2.10
PASSES						
2						
Service						
Naphtha stabiliser (debutaniser)						
Design pressure						
13.0 barg						
Design temperature						
235 C						
Operating P / T						
10.8 / 79 / 203 C						
Shell material						
SA-516-70						
Cladding / lining						
CS (HIC-resistant plate, NACE MR0103)						
Corrosion allowance						
3.0 mm						
Joint efficiency						
1.0 (full RT)						
Shell thickness						
14 mm (by course, bottom up)						
Heads (2:1 SE) top / btm						
16 / 16 mm						
Skirt OD x t x height						
2128 x 8 x 6000 mm						
Anchor bolts						
20 x M30 on 2378 BCD						
Hydrotest (top / btm)						
16.90 / 18.77 barg						
Insulation						
75 mm mineral wool						
Weight empty / oper.						
48 / 54 t						
Weight hydrotest						
111 t						
Wind base V / M						
138 kN / 1992 kNm						
Code						
ASME VIII Div.1, U-stamp; ASCE 7-16						

C-106 NOZZLE SCHEDULE						
MARK	QTY	SIZE	RATING	LOCATION	ORIENT	SERVICE
N1	1	8"	150# RF	EL 123.100	90	Feed (tray 21)
N2	1	20"	150# RF	TOP HEAD	0	Overhead vapour to A-107
N3	1	6"	150# RF	EL 135.350	270	Reflux
N4	1	16"	150# RF	BTM HEAD	0	Reboiler feed to E-117 (thermosyphon)
N5	1	20"	150# RF	EL 111.630	120	Reboiler return from E-117 (thermosyphon)
N6	1	10"	150# RF	EL 107.800	200	Bottoms to P-117A/B
N7	1	8"	150# RF	TOP HEAD	0	PSV-1007 relief (T)
NBA/B	2	2"	150# RF	EL 107.700	300	LT / LG bridle
N9	1	2"	150# RF	TOP HEAD	0	Vent / steam-out
M1	1	24"	150# RF	EL 136.000	0	Manway 24"
M2	1	24"	150# RF	EL 129.205	0	Manway 24"
M3	1	24"	150# RF	EL 123.105	0	Manway 24"
M4	1	24"	150# RF	EL 117.005	0	Manway 24"
M5	1	24"	150# RF	EL 110.930	0	Manway 24"
SECTION						
Top (rectifying)	1-20	FLV	0.064	u flood m/s	D calc m	D sel. m
Bottom (stripping)	21-38	FLV	0.210	0.63	2.81	3.10
PASSES						
2						
Service						
Naphtha splitter						
Design pressure						
3.5 barg						
Design temperature						
185 C						
Operating P / T						
1.1 / 84 / 152 C						
Shell material						
SA-516-70						
Cladding / lining						
CS						
Corrosion allowance						
3.0 mm						
Joint efficiency						
1.0 (full RT)						
Shell thickness						
12 mm (by course, bottom up)						
Heads (2:1 SE) top / btm						
12 / 12 mm						
Skirt OD x t x height						
3124 x 12 x 7500 mm						
Anchor bolts						
24 x M42 on 3374 BCD						
Hydrotest (top / btm)						
4.55 / 7.60 barg						
Insulation						
75 mm mineral wool						
Weight empty / oper.						
109 / 130 t						
Weight hydrotest						
334 t						
Wind base V / M						
294 kN / 6308 kNm						
Code						
ASME VIII Div.1, U-stamp; ASCE 7-16						



C-105 NOZZLE ORIENTATION 1:50
(manways alternate 315 / 45 deg - platform side)



C-106 NOZZLE ORIENTATION 1:50
(manways alternate 315 / 45 deg - platform side)

LEGEND			
	Valve tray with downcomer		Packed bed (structured / grid)
	Liquid distributor		Collector / chimney tray
	Liquid level (LLL/LL/LLL)		Manway (shown on face)
	Nozzle - mark and size		Platform with handrail
	Draw sump		External stiffening ring

- NOTES**
- Dimensions in mm, elevations in m. Grade = EL 100.000.
 - Thicknesses are FEED minimum nominal (CFU-000-ME-CAL-001); fabricator to verify.
 - Nozzle orientations preliminary - to be confirmed by piping study.
 - Platforms 1200 wide, 60 % wrap; caged ladders not shown in full.
 - Datasheet CFU-000-ME-DS-001; weights in data/mech.json.
 - C-105: HIC-resistant plate, PWHT, NACE MR0103 (wet H2S).
 - C-105 reboiler E-116 (kettle), C-106 reboiler E-117 (thermosyphon).

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)
GENERIC REFINERY - GULF COAST		
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT		
COLUMN GENERAL ARRANGEMENT		
C-105 STABILISER & C-106 NAPHTHA SPLITTER		
DISCIPLINE	STAGE	SCALE
MECHANICAL	FEED	1:100
DRAWN	CHECKED	APPROVED
CLAUDE CODE	-	-
DRAWING NUMBER	DATE	
CFU-100-ME-GA-003	2026-10-02	
REV		A